

USER MANUAL

BARCC

Brain Atlas Regional Cell Counter

Version 8.10.000 - Generated September 10, 2026

BARCC is a specialized GUI application for analyzing immunofluorescence images. It enables researchers to overlay brain atlas sections, define regions of interest, perform automated cell counting, and export quantitative results.

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<https://github.com/LaingLab/BARCC>

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1. Introduction

BARCC (Brain Atlas Regional Cell Counter) is a specialized desktop application designed for researchers working with immunofluorescence (IF) microscopy images. It combines powerful image analysis tools with brain atlas registration to enable accurate, reproducible regional cell counting.

Purpose

The software allows users to overlay standardized atlas sections onto experimental TIFF images, define regions of interest through painting or atlas-based selection, apply sophisticated cell detection algorithms, and export quantitative results to Excel for further analysis.

Key Capabilities

- Import TIFF images and multi-page PDF atlas files; browse folders with progress tracking, exclude/include, and reload last count
- Allen Mouse Reference Atlas plates with semi-auto stitch (Reflect / move / rotate hemispheres)
- Practical atlas alignment stack: landmark point pairs, Edge Snap (ICP silhouette), then local border refine
- Crop with optional locked aspect ratio (match TIFF, 1:1, 4:3, 3:2, 16:9, or custom)
- Portable .catlas schematics: save labeled atlas + paint once, apply across channels
- Next Channel workflow keeps atlas and regions while switching fluorescence channels
- Freehand paint regions that register for counting (including on atlas coordinates)
- Blob/DoG detection with Adaptive tiles, peak-quality filters, Dual Settings (Config A/B per region)
- Area Tune and Measure Tune (TP/FP/FN/TN) plus Smart Suggest A/B for parameter recipes
- Manual add/remove/split; Show Cell Mask toggle; save/load cell masks across channels
- Random null cell distributions (optionally stratified by atlas region)
- Axons and Nets: regional intensity with background subtraction and counterstain normalization
- Perineuronal (PNN) shells (2x cell area) and intensity export (true + random)
- Count Cells writes per-image Excel plus an optional combined project spreadsheet ({name}_Counts.xlsx)

NOTE

BARCC is particularly valuable for neuroscience studies involving c-Fos, NeuN, axon/PNN markers, or other signals where regional quantification relative to brain anatomy is required. Multi-channel workflows (e.g. label on DAPI, quantify on other channels) are first-class.

What's New in Version 8.10.000

BARCC 8.10.000 is an atlas-alignment, Dual Settings, and project-counts release on top of v8.09.000 detection tools. Use Dual Settings when one detector cannot serve packed and sparse regions on the same slice; place Allen/PDF plates with Landmarks then Edge Snap then Local Refine; Count Cells can append a combined project spreadsheet.

Practical atlas alignment stack

Atlas alignment is now a three-step stack under the Atlas menu. Use it after Import Allen Atlas (or a PDF plate) and loading the TIFF, typically after a first Fit Atlas to Image.

- Atlas > Align: Landmarks (point pairs)... - Click a feature on the ATLAS (magenta), then the matching point on the TISSUE (cyan). Repeat for 3-6 pairs (ventricle, midline, outer edge). Need at least 2 pairs; 3-6 is recommended. Undo pair / Apply Fit / Cancel live in a small status window. Apply Fit computes a similarity transform (scale + rotation + translation, Umeyama) and moves the whole atlas. Esc cancels. Pairs are stored in image coordinates so they survive zoom.
- Atlas > Align: Edge Snap... - Opens a preview/apply dialog that snaps the atlas silhouette to the tissue outline (iterative closest-point / chamfer search). Preview first, then Apply; Restore undoes a preview. Options include: Refine current pose vs From scratch (full rotation search); Allow Translate / Rotate / Scale; Matching (partial overlap); Auto tissue polarity or Force invert (dark tissue); Tissue tightness; include holes (ventricles); use zone outlines vs ink; Per-region only (selected Atlas Manager zone); Flip atlas left-right; keep landmark pairs as a prior after Landmarks. Does not bake rasters until Apply.
- Atlas > Align: Local Refine (guide)... - Step 3 after global pose is close. Turns Border drag resize ON, expands the Atlas Manager ribbon, and walks you through: select a labeled region, optional per-region Edge Snap, grab the red border segment, or Move Selected Region, then paint custom zones the atlas cannot match. Save Atlas Schematic (.catlas) when done.

NOTE

Recommended order: Fit Atlas to Image -> Landmarks (3-6 pairs, Apply Fit) -> Edge Snap Preview then Apply (refine mode, after landmarks) -> Local Refine for individual structures. Crop after the plate is roughly on the section so you do not crop away useful atlas. These alignment tools were deferred in v8.09.000 and ship in 8.10.

Crop box aspect lock

When Global Crop is on, the Atlas Manager ribbon shows "Crop box shape (aspect ratio)". Check "Lock aspect ratio while drawing" and pick a Ratio before you drag:

- Match TIFF image - same shape as the loaded slice (default recommendation).
- Square (1:1), 4:3, 3:2, Widescreen (16:9).

- Custom width x height - type W and H, then Apply custom (or press Enter in the box).
- Uncheck the lock for a free-form rectangle.
- How to... explains the same steps. Changing the ratio while a box is pending re-shapes that box. Then Enter / double-click / Apply Crop to commit.

Dual Settings Mode (Config A and Config B)

One detector rarely fits a whole slice when packed nuclei (e.g. PVN) sit next to sparse tissue (e.g. AHA) or when background is mixed. Dual Settings Mode runs two Blob/Adaptive configs on the same image.

- In Mask Settings, check Dual Settings Mode. A second Blob Detection panel (Config B) and Adaptive Detection - Config B appear to the right of Config A. The first time you enable it, Config B is copied from Config A.
- Assign labeled regions in Atlas Manager: select a region and press A or B (ignored while typing in an entry box). Or use the Atlas Manager context commands Use Config A / Use Config B. The region list caption shows the assignment.
- Show Mask and Count Cells then detect with Config A in A-regions and Config B in B-regions, and merge the two label maps (unique cell IDs). Unassigned regions default to A.
- Smart Suggest A / Smart Suggest B appear when Dual Settings Mode and "Labeled regions only" are both on. Autotune can target A, B, or both (checkboxes under Autotune).
- Zone A/B assignments are stored with paint bundles so they reload with the schematic.
- Off = one global detector (default).

NOTE

Typical split: assign packed, bright clusters to Config A (denser packing, slightly lower threshold) and dim sparse fields to Config B (higher SNR / bg-relative, sparser packing). Run Smart Suggest A then Smart Suggest B with Labeled regions only checked so each recipe is fit from its own pixels.

Extra blob quality filters and Adaptive region mode

On top of the v8.09 peak-quality gates (local SNR, bg-relative, isotropy, circularity, tissue-edge reject), 8.10 adds filters aimed at folds, knife lines, bubble rims, and packed Fos clusters:

- blob_tissue_margin - reject peaks within N pixels inside the OUTER slice border (bright edge-line FPs). Integer ≥ 0 ; 0 = off. Try 6-12 (15-25 for thick glow).
- blob_max_elongation - max major/minor axis ratio (1 = circle). Rejects thin ridges (folds, fibers, cut-edge line). Default 3.0; 0 = off.
- blob_ridge_reject / blob_ridge_thresh - Hessian ridge test. This is what removes the white midline artifact. 1 = on; thresh typical 0.35-0.55.
- blob_cavity_rim - kill zone around AIR BUBBLES (compact bites in the section edge). Does not clear PVN/SCN next to the 3rd ventricle except peaks actually in the lumen. 0 = off; try 16-32.
- blob_chain_reject - 1-D line/ring suppression (ventricle wall, bubble rim, fold). 0 = off, 1 = default, 2-3 =

stronger. Packed 2-D clusters are kept.

- blob_cluster_recover / blob_seed_snr / blob_recover_factor - two-tier placement: bright nuclei seed a cluster and dim neighbors are kept; isolated sparse cells stay if they look like cells; crowded speckle without a seed is dropped.
- adaptive_region_mode - 0 = classic square tiles over the whole image; 1 = detect only inside painted/atlas structures, each with its own threshold/SNR (so lcl vs lcr are independent). Unlabeled tissue is not labeled. Requires Adaptive on and zones. Then Show Mask.
- Labeled regions only (Blob Detection checkbox) - Show Mask / Count Cells apply detections only inside painted or atlas regions, with a local threshold per region. Separate from Smart Suggest's "Labeled regions only" checkbox.
- Autotune extras: Denser packing, Sparser packing, Background higher (in addition to More/Less cells).

Project counts spreadsheet

File > Select Project Output Directory asks for a folder, then a project name. BARCC writes a combined workbook:

- Path: {chosen folder}/{project name}_Counts.xlsx (sheet "Project Counts").
- Layout: column File plus one column per unique structure name; each later row is one image. Re-counting the same file replaces that row. New structures add columns (never duplicated).
- Count Cells still also writes the per-image workbook under output/counts/ (Cell Counts + Detection Parameters) and the _masked.tif overlay.
- If the .xlsx is open in Excel (locked), BARCC falls back to a CSV next to it.
- Legacy combined files named BARCC_project_counts / *_project_counts are still recognized.

File Browser: exclude, include, reload last count

- Right-click a TIFF: Exclude removes it from Next Uncounted / progress as skipped (persisted under ~/.barc/). Include puts it back.
- Reload last count (paint, mask, config) - restores the last Count Cells artifacts for that image: detection JSON, paint/.barccpaint, cell mask, and counts table when present. Also Cell > Reload Last Count Session... for the open TIFF.
- After reload, the cell mask is locked (Count Cells will not re-detect). Use Cell > Show Mask if you want a new detection.
- View > Show Cell Mask toggles the red detection rings without re-running detection. Rings now survive zoom and pan (persistent overlay layer).

Smarter Add Cell brush

Add Cell no longer only stamps a disk. Click a nucleus: BARCC traces a cell-like blob at that peak, then the brush size (1-10) tightens or expands the fill (1 = tight nucleus, 10 = more aggressive halo, 4 = traced shape

as-is). Right-click still brush-erases. Detection rings stay visible under the edit.

The rest of this manual (chapters 5, 6, 8, 10, 12, 13) describes these tools in the daily workflow, not only in this changelog.

What's New in Version 8.09.000

BARCC 8.09.000 is the published detection and Mask Settings release: Adaptive detection, Area Tune, Measure Tune (TP/FP/FN/TN), smarter Smart Suggest, peak quality filters, and mask-edit UX fixes. Python 3.14 is the recommended runtime.

Adaptive detection (Blob / DoG overlay)

- Adaptive is a checkbox on Blob / LoG or DoG (not a separate radio method). Disabled under Watershed.
- Tile-local thresholds, optional dual-pass (sensitive + strict), density-aware packing.
- Parameters: tile size, tile overlap, sensitivity, packing, dual-pass.
- Autotune More/Less Cells nudges adaptive knobs when Adaptive is on.
- Inactive method panels are dimmed and locked so you cannot edit Watershed knobs while Blob is selected (and vice versa).

Peak quality filters

- Local SNR - (mean core - mean surround) / std surround. Raise (1.5-3.5) to reject high-background texture that is not locally brighter.
- bg relative - peak intensity minus local median (normalized 0-1). Try 0.08-0.18 on high background.
- Isotropy - radial symmetry around the peak; rejects edge-of-tissue and fiber detections bright on one side only (try 0.4-0.55).
- Circularity - local shape; raise 0.55-0.75 to reject peanuts / merged doublets. Values above 1 are treated as 0.80.
- Tissue-edge reject - outer ring partly outside the tissue (near-black). Does not treat a pure dark-field image as "outside". Use with blob_tissue_margin for bright edge lines.

Area Tune

In Mask Settings, click Area Tune. Draw 10 independent diameter lines (one per representative cell). BARCC sets blob_min_area / blob_max_area to 0.7x-1.5x mean area ($\pi * r^2$). Results stay for the session so Smart Suggest can prefer those bounds. Measure Tune does not overwrite Area Tune size bounds when both have been used.

Measure Tune (TP / FP / FN / TN)

After Show Mask / Smart Suggest, click Measure Tune (TP/FP/FN/TN) in Mask Settings. Label the current mask:

- TP (green) - real cell correctly detected.
- FP (orange) - false mark.
- FN (blue) - missed cell.
- TN (gray) - true empty background.
- Precision pass: FP + TN only (≥ 2 should-not) - tighten threshold / SNR / quality.
- Recall pass: TP + FN only (≥ 2 should-detect) - recover missed cells.
- Full pass: both sides. Detection rings stay visible while labeling. Apply uses a progress dialog.
- Performance: local-patch LoG (no full-frame LoG storm on large TIFFs). Labels feed Smart Suggest.

Smart Suggest upgrades (8.09)

- Regional diagnosis (bright vs dark tiles: peaks vs detections).
- Named recipes: mixed_both, high_bg_fp, low_bg_fn, recover_clusters, global over/under.
- Joint suggestions (Adaptive + dual-pass + SNR + threshold + packing together).
- Can turn Adaptive on when indicators support it.
- Trajectory memory after Apply (e.g. ease SNR after a strict high-BG pass).
- Area Tune / Measure Tune results preferred over pure LoG-derived sizes when present.
- Everything still runs 100% on your computer.

Mask edit UX (8.09)

- Remove-cell brush paint is yellow/gold (not red or cyan). Add stays red.
- Detection mask stays visible under add/remove paint (rings + brush composite; works while zooming).

See release-notes-v8.09.000.md in the repository root. Atlas alignment, Dual Settings, and project counts that were deferred from 8.09 are in Version 8.10.000 (preceding chapter).

What's New in Version 8.08.000

BARCC 8.08.000 is a major multi-channel atlas and analysis release: Allen atlas integration, portable .catlas files, cross-channel cell masks, axon/PNN intensity with corrections, random null distributions, and perineuronal shell measurement.

Allen Mouse Atlas & hemispheres

- Atlas > Import Allen Atlas: plate browser with Nissl reference strip and borders-only movable overlay.
- Semi-auto stitch editor: Reflect right-to-left, move/rotate halves, then Load into BARCC.
- Fit Atlas to Image, crop/move/scale/rotate with model-space placement (stable under zoom).
- After Reflect / bilateral stitch, structure names use _r and _l suffixes (e.g. V2M_r / V2M_l) so left and right are distinct in Atlas Manager and Count Cells.
- Download Full Allen Atlas option for offline/cache use.

.catlas schematic (multi-channel atlas)

- Atlas / File > Save Atlas Schematic and Load Atlas Schematic save a lossless .catlas package: zone mask, structure drawings, painted regions, Atlas Manager names, and placement (img_x/img_y).
- Label on counterstain (e.g. DAPI), save .catlas, load onto other channels of the same section.
- When background size changes between save and load, layers and offsets scale together (avoids drift).
- Legacy .atlas files still open; new saves use the .catlas extension.

Next Channel & Clear Atlas

- File / File Browser / Atlas Manager > Next Channel loads a new TIFF while keeping atlas drawings, zones, names, paint, and placement.
- Normal Import TIFF / Next Image fully clears atlas (no ghost double overlays).
- Clear Atlas removes overlay and labeled regions but keeps the TIFF.

Paint on atlas & menus

- Painted regions use atlas coordinates when an atlas is loaded; named paint merges into the zone mask for Count Cells without wiping Allen structures.
- Mask menu renamed Cell: detection tools plus Counting submenu (Count Cells, Show Zone Labels & Counts).
- Axons and Nets menu for intensity, counterstain normalization, and perineuronal tools.

Intensity, cell masks, random null, PNN

- Measure Region Intensities with optional Xth-percentile background subtraction and counterstain normalization (file from Counterstain Normalization Measurement).
- Exports include Pre_Correction and Post_Correction mean/median columns; Excel under output/intensities/.
- Save / Load Cell Mask (.barccmask + PNG) across channels; loaded mask locks Count Cells (no re-detect until Show Mask).
- Generate Random Cell Mask: same count as ground truth, random XY; stratified by atlas region when a .catlas/atlas is loaded (red = true, cyan = random).
- Draw Perineuronal Masks: shell from cell edge out to a disk of 2x cell area; also for random cells if present.
- Measure Perineuronal Intensity: by-structure mean/SEM and median/SEM (true and random); per-cell tables with area + PNN intensity.
- Output folder organized by feature: output/counts/, intensities/, pnn/, atlas/, cell_masks/, paint/, flattened/ (legacy flat files still detected).

See release-notes-v8.08.000.md in the repository root for the full changelog. Count Cells still counts only the ground-truth cell mask (not the random null mask).

What's New in Version 8.07.000

BARCC 8.07.000 improved Save Flattened Image so exports match what you see after painting and counting.

- Flattened export now composites TIFF base + yellow zone fills + black paint boundaries + red cell outline rings (when available).
- Default name {tiff}_flattened.tif under output/flattened/ when possible.
- Hardened overlay steps so partial failures still save a usable base image.

What's New in Version 8.06.000

BARCC 8.06.000 is a stability and usability release focused on Count Cells reliability and the Zone Labels & Counts viewer.

- Show Zone Labels & Counts now works: Cell > "Show Zone Labels & Counts" opens a dedicated table window listing every zone name and its cell count for the current file. Data comes from the latest Count Cells run, saved .xlsx/.csv results in the TIFF folder, or defined zone names (with dashes if not yet counted). A total row appears when numeric counts are available. The window refreshes after counting and when switching files in the left browser.
- Fixed hard crash at end of Count Cells on Windows: Saving the automatic `\_masked.tif` overlay used `compression='tiff\_deflate'`, which can segfault some Pillow/libtiff builds. The masked image is now saved as a standard uncompressed TIFF (same filename, stable on all platforms).
- Fixed Count Cells loading full-resolution TIFFs into memory when the canvas was not yet laid out (1x1 pixel at startup), which caused extreme slowdowns and apparent crashes on large microscopy frames. TIFFs now scale to fit the viewer window using screen/window dimensions as a fallback.
- Count Cells no longer calls the full Stop Paint path mid-count (which triggered `save_state`, `save_paint`, menu corruption, and extra redraws). A lightweight finalize step commits paint strokes to zones without UI side effects.
- Faster, safer per-zone counting: when a final cell mask is already computed, the redundant second watershed pass is skipped (connected components are used instead).
- Additional guards: try/except around the full count pipeline with user-visible error dialogs; 2D mask shape validation; progress dialog always closes in a finally block.

Together these changes make Count Cells complete reliably on typical Windows installations and give users an immediate tabular summary of regional results.

What's New in Version 8.05.000

BARCC 8.05.000 fixed critical issues when loading painted region bundles (.barccpaint) onto TIFFs and running Count Cells.

- Fixed "ufunc 'less'" error when loading .barccpaint bundles with named painted regions (JSON string zone IDs vs uint8 mask labels).
- Fixed silent Count Cells failures after bundle load: numpy edge cases in marker drawing, progress dialog timing vs results popups, and missing feedback when save paths were unset.
- Zone ID keys normalized to int on every load path; manual cell-edit masks cleared on new TIFF load.

What's New in Version 8.04.000

BARCC 8.04.000 focuses on major improvements to the Paint tool workflow, undo reliability for custom regions, and precision editing of painted region boundaries.

- Full repeated undo support for painted regions: Each individual paint stroke (mouse-down to mouse-up group) now creates its own undo checkpoint at the start of drawing. Naming a painted region, Stop Paint (auto-naming), Count Cells (force conversion), border/edge deformations, and Move Selected on painted zones are all fully undoable (up to 40 levels in the bounded history).
- Undo button: Prominent <- Undo button added to the Atlas Manager ribbon header (always visible when ribbon is shown). Also available via Edit > Undo (with Ctrl+Z accelerator shown). Keyboard Ctrl+Z continues to work. The ribbon list/header and visual elements (including paint strokes) update immediately after each undo step.
- Painted region border/edge expansion with deferred commit: When "Border drag resize enabled" is active and a painted region is selected, live dragging near the edge (or using the red local edge segment) updates the yellow/orange highlighted zone mask in real time for preview. The original black drawn boundary line stays in its previous position during the drag (providing a clear before/after view).
- Press **Enter** (or keypad Enter) after adjusting the border to commit: The current mask contour is extracted, the stored painted zone outline points are updated to the final deformed shape, ``_rebuild_paint_layer_from_data()`` is called to re-rasterize a clean black boundary line from the new points (with proper caps and joints), and a full ``show_page()`` redraws it. This "bakes" the expanded painted region and refits the visible black outline exactly to the new mask shape. The mask/zone data was already updated live; Enter finalizes the visual boundary.
- "Border drag resize enabled" checkbox now starts **unchecked** by default (in the Atlas Manager ribbon under the selected region tools). You must explicitly check it after selecting a region before edge or one-sided border drag tools will activate. This prevents accidental activation of the advanced editing mode.
- Paint mode indicator: When the Paint tool is active (entered via Paint > Start Paint or equivalent), a clear "[Paint] PAINT ON" label appears in bold red in the ribbon header. The main window title also updates to include " - [Paint] PAINT MODE". The indicator returns to "Paint: off" (gray) on Stop Paint, tool switches, or other exits from paint mode. The indicator is visible even if you hide the full ribbon (title bar fallback).
- Edge/border manipulation, "Move Selected Region", and related per-region tools now work fully for painted regions (in addition to atlas regions) using correct coordinate mapping (background/image space vs. atlas page space). Deformations update both the mask (for counting and yellow tint) and, on Enter commit for painted, the black visual boundary.
- Undo stack and state hygiene improvements: Painted region creation (draw + name), deformations, and finalization no longer cause unexpected batch undos or stale banner entries. The Atlas Manager "Labeled Regions" list and header now stay perfectly in sync with the actual zone data and visuals after paint actions and undos. Mask pruning on undo prevents the orphan auto-registration logic from re-adding

removed painted zones.

- All new paint editing and undo operations participate in the existing undo (save_state at the right points) and correctly refresh the ribbon, canvas, and paint_layer.

These changes make precise, iterative editing of custom painted regions (with live mask preview and explicit commit for the visible black boundary) reliable and user-friendly, while keeping the powerful Atlas Manager ribbon workflow intact.

What's New in Version 8.03.000

BARCC 8.03.000 delivers a major new workflow for atlas-based region editing via the Atlas Manager ribbon, plus global quick adjustments, improved mode visibility, and numerous robustness fixes for mixed image+atlas workflows.

- Atlas menu reorganization: "Import Atlas" (and global Crop/Move/Rotate/Scale plus per-region tools) now lives under a dedicated Atlas top-level menu (moved from File menu for better discoverability).
- New Atlas Manager ribbon (collapsible/expandable via header arrow; fully toggleable on/off from View > "Show Atlas Manager Ribbon" check item). The ribbon is the central hub for atlas region work:
 - Header always shows the currently selected region name/ID (or "No region selected").
 - Global tools now implemented as checkboxes (Crop, Move) so the active mode is immediately visible. Checking enables the corresponding click-drag behavior on the canvas (whole-atlas operations); unchecking returns to normal selection/naming.
 - "Move Selected Region" checkbox: when a region is selected (orange tint), click+drag inside it to translate *only* that zone's pixels in the mask. The underlying atlas artwork, other regions, and background image remain fixed. This is ideal for fine local alignment corrections without disturbing global registration.
 - "Border drag resize enabled" checkbox: when checked (and region selected), clicking near the perimeter of the orange/yellow region illuminates a local red edge segment. Re-click the red line to reposition the editable window; drag the red to locally push/pull only that portion of the boundary (linear falloff weights preserve connectivity and overall region integrity). Live shape update of the tinted region during drag via efficient rasterization + partial refresh. Release commits the edit.
 - Persistent red edge highlight survives panning, zooming, page changes, and show_page() calls (unless explicitly toggled off by a short click on an already-illuminated edge).
 - Quick Global Adjust section (new): Rot +5° / -5°, Scale +5% / -5% buttons that apply to the entire current atlas page (base artwork + all zone masks). Plus "Dialogs..." shortcut to the full global Rotate/Scale settings dialogs.
 - Selectable list of all labeled regions on the current atlas page. Clicking any entry selects it for editing (orange tint, ribbon header update, ready for quick adjust, edge drag, or move-selected). List auto-syncs after naming, transforms, page changes, etc.
 - Per-region quick adjust (unchanged from prior work but now clearly labeled "Selected Region Quick Adjust"): Rot +/-5°, Scale +/-5%, and "Dialogs..." for the currently selected region only (centroid-preserving transforms via the existing _apply_transform_to_region machinery).
 - Automatic mutual exclusion between global modes and edge features: enabling the border-drag checkbox deselects global Move (and Crop); enabling global Move or Crop automatically unchecks the border drag. Region "Move Selected" also cooperates with edit mode bindings. This prevents confusing overlapping behaviors and makes the active toolset obvious from the checkbox states.
- Major improvements to per-region atlas editing: individual regions on an imported atlas can now be

selected (canvas click on named yellow area, list, or Atlas menu Select Region tool), then independently rotated, scaled, translated (via the move-selected drag), or locally deformed via the red-edge drag. The rest of the atlas and background stay in place. Transforms update both the visual tint and the underlying label mask used for counting.

- Robustness fixes across the board:
 - Atlas crop now correctly converts canvas rect to model (native) coordinates using `_canvas_to_atlas`, clamps to image bounds, rebases `img_x/img_y` so the cropped content stays visually in place, prunes orphaned zone names, and fully clears stale selection/edge state.
 - Load-order fixes: loading a TIFF/image *after* an atlas no longer leaves stale `selected_zone_id` or `selected_edge_full_contour` that would hijack global Move drags (via the priority checks in `drag_start/drag_move`) or cause `is_near` tests to behave unexpectedly. All image-load paths (`import_tiff` and file-browser `_load_tiff_file`) now perform complete selection/edge/region-mode clears in addition to the zone/mask wipes.
 - Edge grab hit-testing made forgiving: clicks that land on boundary pixels (`zid==0` at exact integer sample) but are near the current selected region's border (or an already-illuminated red) are treated as edge-grab intent instead of falling through to a name prompt.
 - Global Move (`edit_mode`) and per-region features now correctly delegate in the drag handlers so that "Move Selected Region" or edge grab work even if the global Move binding is active (priority checks on `mousedown`; motion delegation).
 - Many additional state hygiene improvements (clears on page change, deselect, import atlas, crop end, etc.) so that ribbon list, orange tint, red edge, and move/edge flags stay consistent.
 - Ribbon checkboxes (global Crop/Move + per-region Move Selected + Border drag) plus the selected region header and list now give immediate, at-a-glance feedback about the current editing context and which tools are armed.
 - All new per-region and global quick adjust operations participate in undo (`save_state`) and correctly update the ribbon list/header after transforms.

These changes make fine-grained, region-by-region correction of atlas registration practical while preserving the ability to do global alignment. The visual distinction between "I'm moving the whole atlas" vs. "I'm only moving/deforming this one region" is now explicit via checkboxes and selection highlighting.

What's New in Version 8.02.002

BARCC 8.02.002 is a patch release that fixes a long-standing source of confusion between the counts the user sees visually on the annotated image/mask and the numbers reported in the spreadsheet.

- Fixed mismatch between per-zone cell counts shown on the mask/image (yellow zone labels like "RegionName\n(NN)" when "Show Zone Labels & Counts" is enabled, plus visible red cell markers in the annotated image) and the numbers in the exported spreadsheet (Cell Counts sheet) or CSV fallback.
- Root cause: The cell-to-zone assignment inside `count_cells_in_zones`` (used for the DataFrame that feeds both the spreadsheet and `last_df``) and the on-screen label positioning in `show_page`` unconditionally applied the atlas overlay offset (`-img_x` / `+ display_img_x`` etc.).
- Painted regions (and the main experimental TIFF background) live in a 0,0-aligned coordinate system matching the cell detection mask. Atlas/PDF zones may come from a differently sized/positioned overlay layer. Any non-zero `img_x`/img_y`` (from panning the atlas, zooming, dragging, alignment, etc.) would shift the lookup, causing cells to be assigned to the wrong zone (or no zone) for counting purposes.
- Visual markers were drawn at the correct positions (using raw centroids), so users would see e.g. 30 red dots inside a painted region but the spreadsheet and labels would report a different number.
- Implemented a robust size-based heuristic:
 - Compare zone mask size to the main `background_image`` / cell mask size.
 - If they match (within tolerance): zones are paint-on-background or standalone TIFF -> use direct cell coordinates (no offset) for accurate assignment and place labels over the background layer at canvas (0,0).
 - If sizes differ (atlas overlay): fall back to the previous `img_x`/img_y`` offset logic to align with the placed atlas layer.
- The same logic was applied to zone label text placement so the yellow (count) overlays appear over the correct regions even when an atlas overlay has a non-zero offset.
- After re-running Count Cells, the per-zone totals in the spreadsheet now exactly match the number of cells the user sees marked inside each region on the image and in the on-screen labels.
- No change to pure atlas workflows, detection logic, manual edits, or cases where no overlay offset is present. Version in exported settings JSON is "8.02.002".

This resolves the final major "numbers don't match what I see" issue for users relying on painted custom regions for quantitative analysis.

What's New in Version 8.02.001

BARCC 8.02.001 is a targeted patch release that resolves the last reported edge cases in the Paint tool for custom regions and improves dialog usability:

- First-paint reliability: Drawing a region, naming it immediately via right-click, and clicking Count Cells now succeeds on the **very first attempt** after loading any image (no more "No Regions Defined" for the initial named zone, with only later zones appearing in the spreadsheet).
- Fixed root cause in the 'img' bake path: The first ``stop_paint()`` (auto-called by Count Cells) + its ``show_page()`` would activate ``atlas_filetype='img'`` (via ``save_paint``). ``load_page_image()`` then unconditionally reset ``zone_names[page]``, ``mask_images[page]``, and ``zone_counters[page]`` the first time ``page_images`` was populated for non-PDF content. This destroyed zones just registered by ``name_painted_region`` -> ``_convert_named_paints_to_zones``. A guard now restricts the destructive per-page init to ``atlas_filetype == 'pdf'`` only; 'img' (baked paint) and 'png' (loaded paint) cases preserve existing zone/mask state.
- Additional conversion hardening (building on 8.02.000 durability work): broadened stroke collection in ``_convert_named_paints_to_zones`` (always tries ``paint_group_data`` model points + canvas + last-resort any paint items for still-named groups), conditional ``dtag`` only after successful retirement in ``name_painted_region``, pre-clear re-try collection inside ``stop_paint``, and an ultimate force-add of lingering data groups before the error guard in ``count_cells``.
- Brightness slider dialog X button fix: The titlebar X (and Alt+F4) on the Brightness Settings window (containing the live brightness scale/slider) now properly closes the dialog. Previously wired to a no-op ``disable_event``. Applied the same ``window.protocol("WM_DELETE_WINDOW", window.destroy)`` pattern (already used by Mask Settings) to Brush Settings, Scale Settings, and Rotate Settings for consistency. Transparent window registration and `<Destroy>` cleanup continue to work. Progress dialogs remain hardened (defensive no-op flag) so early close cannot crash counting/detection.
- All prior v8.02.000 guarantees (immediate naming registers zones, Count auto-stops paint, durable model coordinates, ``binary_fill_holes`` + neighborhood lookup for accurate interiors, retirement to prevent dups, full wipe on every new image load, auto Save Paint Layer into the left File Browser directory, etc.) now apply reliably even to the first painted+named region on a fresh load.

These changes complete the Paint tool reliability story. The workflow "draw, right-click name immediately after the stroke, click Count Cells (without pressing Stop Paint)" is now fully production-ready on the first try.

What's New in Version 8.02.000

BARCC 8.02.000 delivers major stability, accuracy, and reliability improvements focused on the Paint tool workflow for creating custom analysis regions:

- Painted zones that are named immediately after drawing (via right-click) now correctly and reliably register for cell counting. Clicking Count Cells automatically stops paint mode and converts all strokes - both user-named regions and auto-defaulted "Painted Region N" entries.
- Complete prevention of cross-image contamination: every new TIFF loaded (via File Browser or Import) performs a full wipe of all mask_images, zone_names, zone_counters, named_paint_groups, and durable paint geometry data.
- Durable storage of paint stroke geometry using stable model/image coordinates (not view-dependent canvas coords). Hand-drawn regions now survive zooming, panning, and internal canvas refreshes.
- Accurate interior filling of painted structures using `scipy.ndimage.binary_fill_holes` after boundary stroking. Cells that are visibly inside your drawn outlines are now counted correctly instead of only thin boundary pixels.
- Neighborhood search (5x5) around each detected cell centroid when looking up zone membership. This provides tolerance for drawing precision and any remaining micro-imperfections in region fills.
- Critical stability fix: Closing the "Counting Cells" or "Detecting Cells" progress dialog with the X button (or Alt+F4) before the operation finishes no longer crashes the application. All progress UI methods are now fully defensive no-op after early dismissal.
- Elimination of duplicate zone entries (e.g., 6 zones appearing when only 3 were drawn) through proper retirement of processed paint groups after conversion.
- Overall hardening of the entire paint -> named group -> zone mask -> counting pipeline.
- Paint menu reorganization: "Save Paint" moved from File menu to Paint menu and renamed "Save Paint Layer". New "Load Paint" command added.
- Save Paint Layer now auto-saves directly into the folder currently open in the left File Browser (with automatic unique naming to avoid overwrites), matching the auto-export style of Count Cells. No save dialog appears when a working directory is selected.
- Load Paint and Import Paint now default their file dialogs to the current directory shown in the left File Browser for faster workflow.

These changes make the Paint tool (for custom regions outside of atlas PDFs) fully trustworthy and production-ready for quantitative cell counting work.

What's New in Version 8.01.000

BARCC 8.01 introduces major improvements to the cell detection system and parameter tuning workflow:

- New default detection engine based on Laplacian-of-Gaussian (blob_log) blob detection. This method is significantly more robust on typical immunofluorescence images than the previous watershed approach.
- Full set of Blob Detection parameters now exposed in Mask Settings with live preview.
- New "Smart Suggest (Pre-tuning smart settings)" button - a completely local analysis tool that examines your image and current detections and recommends better parameter values (pre-tunes smart settings). No data is ever transmitted.
- Ability to instantly switch between the new Blob method and the legacy Watershed method.
- Autotune buttons now intelligently adapt their adjustments depending on the active detection method.
- When using Add Cell or Remove Cell, the Brush Settings dialog now opens automatically for immediate size control.

These changes dramatically improve the experience of tuning detection parameters for difficult or variable images.

2. Installation & Requirements

System Requirements

- Windows 10 or later (primary supported platform)
- Python 3.14 recommended (conda env barcc314). Python 3.8+ still runs; 3.12 (legacy env barcc) is a fallback
- At least 8 GB RAM recommended for large images (16 GB+ for very large TIFFs)

Image Format and Compatibility Requirements

BARCC is designed primarily for quantitative analysis of immunofluorescence images. The following specifications describe the supported and recommended image characteristics:

Supported File Formats

BARCC supports the following image file formats with varying levels of compatibility:

- TIFF (.tif, .tiff) - Fully supported and strongly recommended. Supports both single-page and multi-page (stacked) TIFF files. Handles 8-bit, 16-bit, and 32-bit floating point data. This is the only format recommended for serious quantitative work.
- PNG - Limited support. May be imported in certain workflows but is not recommended for primary analysis due to potential compression artifacts.
- JPEG / JPG - Not supported. Lossy compression makes these formats unsuitable for cell detection and quantitative analysis.

Bit Depth and Data Types

- 8-bit (uint8) - Supported
- 16-bit (uint16) - Fully supported and recommended for most immunofluorescence work
- 32-bit floating point - Supported (both 0-1 normalized and arbitrary ranges)
- Multi-channel images (RGB, RGBA, etc.) - Accepted but typically converted or reduced internally. Single-channel grayscale images are strongly preferred for best results.

Image Dimensions and Size

BARCC has no hard upper limits on image dimensions, but practical performance considerations apply:

- Recommended maximum: approximately 8,000-10,000 pixels on the longest side for comfortable interactive use.
- Very large images (e.g., > 12,000-15,000 pixels per side) may cause high memory usage, slow

performance during detection, or instability during manual editing.

- File sizes above ~500 MB can lead to noticeable slowdowns in the Mask Settings dialog and when generating visualizations.

Resolution and Physical Scaling

BARCC is strictly pixel-based and does not read or use image resolution metadata (DPI or $\mu\text{m}/\text{pixel}$). All detection parameters (cell size, sigma values, peak intensity, etc.) are expressed in pixels. Users must convert biological size requirements into pixel values based on their specific imaging resolution.

Recommended Image Characteristics

For optimal performance with the current detection system (especially the Blob Detection method):

- File format: Uncompressed or lossless TIFF
- Bit depth: 16-bit or 32-bit floating point
- Background: Relatively dark and uniform
- Cell appearance: Locally brighter than background, roughly circular or elliptical
- Noise level: Moderate to low (use preprocessing denoising when necessary)
- Image size: 1,000-8,000 pixels on the longest side for best interactive experience

NOTE

While BARCC can technically load images outside these recommendations, results may be suboptimal. For best accuracy and performance, prepare images as single-channel TIFFs with good contrast between cells and background.

Installation Steps

1. Clone the repository:

```
git clone https://github.com/LaingLab/BARCC.git
cd BARCC
```

2. Create the Python 3.14 conda environment (recommended) and install dependencies:

```
conda env create -f environment.yml
conda activate barcc314
pip install -r requirements.txt
```

environment.yml names the env barcc314 (Python 3.14). Alternatively: conda create -n barcc314 python=3.14 pip -y, then activate and pip install -r requirements.txt. requirements.txt includes openpyxl and xlsxwriter for Excel export. If those engines are missing, Count Cells falls back to CSV.

3. Launch the application (Windows):

- Double-click Application/Launch_BARCC.bat (tries barcc314 pythonw, then the legacy barcc 3.12 env,

then py -3.14).

- Or double-click Application/BARCC.Ink if you created a shortcut.
- Or from Anaconda Prompt: conda activate barcc314, then cd Application, then python barcc.py.

Generic launch:

```
cd Application
python barcc.py
```

3. Getting Started

Upon launching BARCC you are presented with a large central canvas area and a menu bar across the top. The typical workflow follows these steps:

- Import a TIFF (File > Import TIFF or File Browser). Optionally File > Select Project Output Directory for the combined counts workbook
- Import an atlas (Atlas > PDF or Allen Atlas + stitch; optionally save .catlas)
- Align: Fit Atlas to Image, then Landmarks (3-6 pairs) -> Edge Snap Preview/Apply -> Local Refine / Crop (lock aspect if desired)
- Define regions via atlas click, Paint, or Load Atlas Schematic. For Dual Settings, assign regions A or B
- Optional multi-channel: Next Channel (keep atlas) or load .catlas / cell mask
- Configure detection (Cell > Show Mask Settings): Adaptive, Area Tune, Measure Tune, Smart Suggest; edit/save cell masks as needed
- Count Cells and/or Axons and Nets intensity / PNN measurement
- Results auto-save under the image output/<feature>/ folders; project workbook updates if you chose a project directory

4. User Interface Overview

The main interface consists of a left file browser pane and a large central image canvas. The left pane lets you select a folder and browse all TIFF images within it. Double-clicking any file loads it as the active image. A status column shows which images have already been counted (matching results under output/counts/ or legacy .csv/.xlsx). Right-click a file to Exclude / Include it from batch navigation or to Reload last count.

All other functionality is accessed through the top menu bar and the Atlas Manager ribbon. View > Show Atlas Manager Ribbon and View > Show Cell Mask control the ribbon and the persistent red detection rings. The interface keeps as much screen space as possible for the image.

Main Canvas

The large central area displays your experimental image with the atlas overlay on top (when loaded). You can pan using the scrollbars or Alt+drag. Use the mouse wheel to zoom in and out. Zoom is centered on the mouse cursor and keeps painted regions, atlas overlays, and masks perfectly aligned with the background image at all zoom levels.

Menus

File, Edit, Atlas, Paint, Cell, Axons and Nets, View, and related menus provide access to all features. The former Mask menu is now Cell (detection, mask edit, cell masks, Count Cells, Reload Last Count Session). Atlas includes Import, Fit, Landmarks, Edge Snap, Local Refine, schematic save/load, and per-region tools. File includes Select Project Output Directory and Next Channel. View includes Show Atlas Manager Ribbon and Show Cell Mask. Many operations open auxiliary dialogs for parameter adjustment.

5. File Menu & Multi-Channel Workflow

Import TIFF

Loads the primary experimental image. Supported formats include single and multi-page TIFFs. After import, the image is automatically scaled to fit the window while preserving aspect ratio. Importing a new TIFF clears atlas and zone state (use Next Channel if you want to keep the atlas).

Next Channel (keep atlas)

File > Next Channel, the File Browser button "Next Channel (keep atlas)", or the Atlas Manager ribbon button opens a file dialog for another TIFF (e.g. the next fluorescence channel). Atlas drawings, zone masks, Atlas Manager names, painted regions, and placement offsets are preserved. Cell detection masks are cleared - reload a saved cell mask if you need the same detections. Autosave of counts/paint for the channel you leave is attempted when possible.

Save / Load Atlas Schematic (.catlas)

File or Atlas > Save Atlas Schematic writes a portable .catlas file under the image's output/atlas/ folder by default. Load Atlas Schematic applies that schematic onto the currently open TIFF without re-importing the Allen plate. Ideal workflow: label on DAPI, save .catlas, Next Channel to axon/PNN channels, Load Atlas Schematic if needed (or keep atlas via Next Channel).

Import Atlas (PDF) - under Atlas menu

Opens a PDF file containing brain atlas plates. BARCC uses PyMuPDF to render individual pages at high quality. You can navigate between pages of multi-plate PDFs. Allen Mouse Atlas import and stitch live under Atlas as well (see chapter 6).

Split Tiff

Utility for breaking apart stacked (multi-page) TIFF files into individual images. Useful when your source data contains multiple sections in one file.

Select Project Output Directory

File > Select Project Output Directory chooses where the combined project counts spreadsheet is written. BARCC prompts for a folder, then a project name. The workbook is:

```
{folder}/{project name}_Counts.xlsx
```

Sheet name: "Project Counts". Row 1 is File plus unique structure names; each later row is one image.

Count Cells appends or updates the current TIFF's row (re-counting the same file replaces that row; new structures add columns, never duplicates). Per-image files under output/counts/ are still written. If the combined .xlsx is open in Excel, BARCC writes a CSV next to it instead. The choice is remembered in UI prefs (~/.barc/).

NOTE

Set the project directory once at the start of a cohort. Then Count Cells on each TIFF (or walk Next Uncounted) and open the combined workbook when the folder is done. Do not keep the .xlsx open in Excel while counting if you want live xlsx updates.

File Browser (Left Pane)

BARCC v8.01 introduced a dedicated file manager pane on the left side of the main window. This makes it much easier to work with folders containing many TIFF images.

Click the "Select Folder" button at the top of the left pane to choose a directory. BARCC will scan the folder and display all .tif and .tiff files in a list. Double-click any file in the list to load it as the active working image.

A status column shows which images have already been processed (checkmark when matching Count Cells results exist under output/counts/ or as legacy .csv/.xlsx). This is useful for tracking which images in a large dataset have already been counted.

Right-click a TIFF for a context menu:

- Exclude - skip this image in Next Uncounted / remaining progress (stored in ~/.barc/). The row stays visible but is treated as out of the batch.
- Include - put an excluded image back into the batch.
- Reload last count (paint, mask, config) - enabled when Count Cells has already saved artifacts for that file. Loads the TIFF if needed, then restores detection JSON, paint/.barccpaint, cell mask, and the counts table when present. The restored cell mask is locked until you Cell > Show Mask.

The same reload is also Cell > Reload Last Count Session... for the currently open TIFF (or the File Browser selection).

The Refresh button rescans the current folder. This is useful if you add or remove files while BARCC is running.

NOTE

The File Browser works independently of the traditional "File > Import TIFF" menu. You can still use the menu for one-off files, but the left pane is much faster when working with a whole folder of images. Exclude is for images you do not want in this project's remaining count (bad section, duplicate, etc.) without deleting the file.

Save Flattened Image

Exports a composite of the TIFF, yellow zone fills, paint boundaries, and (when available) red cell outline

rings as a TIFF (typically {name}_flattened.tif under output/flattened/). Useful for figure preparation and session records. Also used by autosave on image switch.

Previous / Next Image & Next Uncounted

Navigate the File Browser list (Ctrl+Left / Ctrl+Right / Ctrl+Shift+Right). These load a new section and clear atlas/zones unless you use Next Channel. Progress checkmarks reflect counted artifacts under output/counts/ (legacy flat files in output/ are still detected).

6. Working with Atlas Sections

Accurate alignment of the atlas to your experimental image is critical for meaningful regional analysis. BARCC supports PDF atlas plates and Allen Mouse CCF plates, with tools centered on the Atlas Manager ribbon for global and per-region control.

Import Allen Atlas & stitch editor

Atlas > Import Allen Atlas opens a plate browser (coronal/sagittal). Preview Nissl + structure outlines, then either Load as drawn or Open stitch editor. The stitch editor Reflects the right-hemisphere Allen drawing to the left, then lets you move/rotate left, right, or both before Load into BARCC. Atlas > Download Full Allen Atlas pre-caches plates for offline use.

After bilateral Reflect/stitch, structure names include hemisphere tags: acronym_r and acronym_l (e.g. V2M_r / V2M_l). Left and right get separate zone IDs for selection and counting.

Fit Atlas to Image, Crop, Clear Atlas

- Fit Atlas to Image: resizes the atlas (and mask/borders) to the TIFF size and aligns top-left corners. Use this first, then Landmarks / Edge Snap.
- Crop: turn on the Global Crop checkbox, set Crop box shape (see below), drag a rectangle; incomplete edge structures can be pruned. Enter / double-click / Apply Crop commits.
- Clear Atlas: removes drawings and Atlas Manager regions while keeping the loaded TIFF.

Crop box shape (aspect ratio)

With Crop checked, the Atlas Manager ribbon shows "Crop box shape (aspect ratio)". Set this before dragging:

- Lock aspect ratio while drawing - checked: the rectangle keeps the selected ratio; unchecked: free-form.
- Ratio dropdown: Match TIFF image (same shape as the slice), Square (1:1), 4:3, 3:2, Widescreen (16:9), or Custom width x height.
- Custom: type W and H, then Apply custom or press Enter in the box. Custom entries are enabled only when lock is on and Custom is selected.
- How to... opens the same instructions. Changing the ratio while a box is pending re-shapes that box.
- Status text on the crop hint shows [lock 16:9], [lock = TIFF ...], or [free aspect].

Practical alignment stack (Landmarks, Edge Snap, Local Refine)

After Fit Atlas to Image, use the Atlas menu alignment tools in order. They are designed as a stack: global similarity from point pairs, optional silhouette snap, then per-structure cleanup.

Step 1 - Align: Landmarks (point pairs)

Atlas > Align: Landmarks (point pairs)... exits Crop/Move/Measure Tune/Area Tune, binds the canvas, and opens a Landmark Align status window (always on top).

- For each pair: click a point on the ATLAS (line/feature, magenta marker), then the matching point on the TISSUE (cyan).
- Use 3-6 pairs (ventricle, midline, outer edge). Apply Fit requires at least 2 pairs.
- Undo pair removes the last pair. Apply Fit runs a similarity transform (scale + rotation + translation) and moves the whole atlas. Cancel or Esc exits without applying.
- Pairs are stored in image coordinates so they survive zoom. After a successful fit they can be used as a prior in Edge Snap.

Step 2 - Align: Edge Snap

Atlas > Align: Edge Snap... opens a dialog: "Snap atlas outline to tissue". Preview first, then Apply. Nothing is baked into the atlas rasters until Apply. Restore returns to the captured pose.

- Search: Refine current pose (+/-12 deg, scale +/-8%, small shift) - use after Landmarks. From scratch - full rotation search when the atlas is still far off.
- Allow Translate / Rotate / Scale - constrain which degrees of freedom the search may change.
- Matching: partial overlap is useful when the atlas plate is a hemisphere or the tissue is cropped.
- Auto tissue polarity (recommended) or Force invert (dark tissue) if the section is dark-field.
- Tissue tightness - how aggressively the tissue mask is eroded/opened before taking the outer contour.
- Optional: include holes (ventricles), use zone outlines instead of ink, Per-region only (selected Atlas Manager zone), Flip atlas left-right, keep landmark pairs as a prior.
- Preview draws the proposed silhouette overlay. Quality text reports the residual. Apply commits; Restore / dialog close without Apply leaves placement unchanged.

Step 3 - Align: Local Refine (guide)

Atlas > Align: Local Refine (guide)... assumes global pose is already close. It turns Border drag resize ON, expands the Atlas Manager ribbon if collapsed, and shows the remaining checklist:

- Click a labeled region in Atlas Manager (or Select Region).
- Optional: Edge Snap with Per-region only to ICP that one zone.
- Grab the red border segment and drag to match tissue, or Move Selected Region to translate one zone.
- Paint custom zones for anatomy the atlas cannot match.
- Save Atlas Schematic (.catlas) when done.

NOTE

Do not start with Local Refine on a badly placed plate - Landmarks and Edge Snap move the whole atlas. Local Refine is for leftover mismatches on individual structures.

The Atlas Manager Ribbon

The ribbon (View > "Show Atlas Manager Ribbon" to toggle visibility) is the primary interface for working with atlas regions. It is collapsible via the >/v arrow in the header.

- Header always displays the currently selected region (name and ID) or "No region selected".
- Global tools appear as checkboxes (Crop, Move) - their checked state shows at a glance whether whole-atlas click-drag modes are active.
- "Move Selected Region" checkbox: enables interior drag of the orange-tinted selected region to shift *only* its mask pixels (everything else stays fixed).
- "Border drag resize enabled" checkbox: arms edge-grab mode for the selected region.
- Global Quick Adjust: one-click Rot +/-5° and Scale +/-5% for the entire current atlas page (base + all masks), plus quick access to the full Rotate/Scale dialogs.
- Selectable list of every labeled region on the current page. Click any entry to select it for editing (orange highlight appears, ribbon header updates). When Dual Settings Mode is on, the list caption mentions A/B: select a region and press A or B (or Use Config A / Use Config B) to assign which detector config that zone uses.
- Selected Region Quick Adjust: the same +/- rotate and scale buttons, but applied only to the currently selected region (centroid-preserving).
- Crop box shape (aspect ratio) panel when Global Crop is on (lock + ratio presets + custom W x H).
- All operations are undoable and immediately reflected in the ribbon list and on-screen tints.

NOTE

The ribbon automatically stays in sync after naming, transforms, page changes, crops, etc. Checkboxes enforce mutual exclusion (e.g., turning on border drag automatically unchecks global Move and vice versa) so you always know which tool family is armed.

Selecting Regions for Editing

Click a yellow (named) or orange (selected) area directly on the atlas canvas, or use the list in the ribbon, or Atlas menu > "Select Region" (temporarily rebinds clicks to pick a zone). The selected region receives an orange tint overlay and becomes the target for quick adjust, edge drag, and move-selected.

Canvas clicks on already-named regions now intelligently autoselect them into the ribbon (instead of re-prompting for a name). Clicks near the perimeter (even on boundary pixels) are treated as edge-grab intent when the border checkbox is on. With Dual Settings Mode on, press A or B after selecting a region to assign Config A or Config B (the list caption reminds you).

Global Quick Adjust (new in 8.03)

The ribbon now offers the same style of quick +/- buttons for the *entire* atlas page that were previously available only for individual selected regions:

- Rot +5° / Rot -5° - rotates the full rendered atlas artwork and all zone masks for the current page (expand=True semantics, size may grow).
- Scale +5% / Scale -5% - uniform scaling of the entire page content.
- "Dialogs..." shortcut to the full global Rotate and Scale settings dialogs (with numeric entry and axis-specific scaling).

These are the global equivalents of the per-region quick adjust. They affect every region on the page plus the underlying artwork; use them for coarse global corrections before fine per-region work.

Per-Region Editing & Quick Adjust

Once a region is selected (orange), you can:

- Use the Selected Region Quick Adjust buttons (Rot +/-5°, Scale +/-5%) - these transform only the chosen zone's mask while keeping its centroid roughly in place. The rest of the atlas and background are untouched.
- Open full per-region Rotate/Scale dialogs via the "Dialogs..." button (or Atlas menu).
- Enable "Move Selected Region" and drag inside the orange area to translate only that zone (see below).
- Enable "Border drag resize" for local boundary editing (see below).

Move Selected Region (translate only one zone)

With a region selected and the "Move Selected Region" checkbox checked, click and drag inside the orange area. Only the pixels belonging to that zone in the label mask are shifted; the atlas artwork itself, other zones, and the background image remain exactly where they are. This is perfect for nudging a single misaligned anatomical region without disturbing your global registration.

The checkbox and the global Move checkbox are coordinated so you can easily switch between moving the whole atlas layer versus moving just the current region.

Edge Grab & Local Deformation (expand/shrink from the border)

With "Border drag resize enabled" checked and a region selected:

- Click near the perimeter of the orange/yellow region. A local segment of the boundary is highlighted in bright red.
- The red segment is a movable "handle" centered on your click point (using a sliding window along the contour with falloff weighting).
- Click again on an already-red edge to re-center the editable window at that exact location.
- Drag the red line outward or inward. Only the vertices inside the local window move, with linear falloff so the deformation blends smoothly into the rest of the boundary. The rest of the region (and the rest of the atlas) stays fixed.

- You get live visual feedback: the tinted region shape updates in real time as you drag.
- Release the mouse to commit the new shape to the mask (used for counting) and the display.
- A short click (no drag) on a red edge toggles the highlight off.

The red highlight is persistent across pans, zooms, and show_page refreshes until you explicitly dismiss it. This gives you precise, one-sided control over region boundaries without having to repaint or globally scale/rotate.

Global Crop & Move (checkbox style)

The Global Crop and Move tools are now checkboxes in the ribbon (and check items in the Atlas menu). When checked, they rebind canvas clicks for whole-atlas operations (crop rectangle or drag-to-move the entire overlay layer via `img_x/img_y`). Their checked state gives clear feedback that you are in a global mode rather than a per-region editing mode. Unchecking (or enabling an edge feature) restores normal click behavior (naming / region selection / edge grab).

NOTE

Best practice for atlas work: Use global Crop/Move/Quick Rotate/Scale first for rough alignment of the whole plate, then switch to per-region tools (list or canvas selection + quick adjust + edge drag + move-selected) for fine anatomical corrections. The checkboxes and orange selection tint make the current context obvious.

Move Atlas (legacy / still available)

The classic click-and-drag of the atlas overlay (when the global Move checkbox is checked) repositions the entire atlas layer relative to the background. This remains the primary tool for coarse global alignment.

Rotate / Scale (global)

Use either the ribbon Global Quick Adjust buttons for small increments or the full dialogs (Atlas menu or ribbon "Dialogs..." under Global Quick Adjust). These affect the rendered atlas artwork and all zone masks on the current page. After a global rotate that expands the canvas, the layer offset is automatically handled on the next redraw.

Brightness & Contrast Adjustments

The atlas rendering can be lightened or darkened independently of the experimental image to improve visibility of boundaries during alignment.

NOTE

Best practice: Identify 3-6 reliable anatomical landmarks (ventricles, midline, major fiber tracts, cortical boundaries) with Align: Landmarks, optionally Edge Snap, then use per-region tools (Local Refine / border drag / Move Selected) to tweak individual structures without disturbing the rest of the plate.

7. Paint Tools for Regions of Interest

When the standard atlas regions do not match your experimental needs, the Paint tools allow completely custom region definition.

Start / Stop Paint

Activates freehand drawing mode. Draw directly on the image with the mouse.

Pen vs Eraser

Switch between adding area (Pen) and removing area (Eraser). Brush size is adjustable via the Brushsize dialog.

Load / Save Paint Layer

Painted regions can be saved and reloaded in future sessions, enabling consistent analysis across multiple images or experiments.

The Paint menu now contains dedicated **Load Paint** and **Save Paint Layer** commands. When a folder is open in the left File Browser, Save Paint Layer automatically saves the current paint layer directly into that folder as `{image}_paint.png` (or with numbered suffixes to avoid overwriting). The left file list is refreshed automatically after saving. Both Load Paint and Import Paint default their file selection dialogs to the current working directory shown in the left File Browser.

Labeling Painted Regions

While in Paint mode, right-click any painted stroke to name it. BARCC treats each continuous drawing action (mouse down through release) as one structural boundary. Naming converts the group into a zone immediately for Atlas Manager and Count Cells. Stop Paint and Count Cells also auto-name remaining strokes as "Painted Region N".

- The connected group highlights in yellow when selected for naming.
- With an atlas loaded, paint uses atlas model coordinates and merges into the existing zone mask without overwriting other structure IDs.
- Named painted regions participate in cell counting and intensity export like atlas regions.

NOTE

Prefer closed loops so interiors fill correctly (flood fill + binary_fill_holes). After naming, the region appears in the Atlas Manager list.

8. Cell Detection, Masks & Editing

Cell detection tools live under the Cell menu (formerly Mask). Show Mask Settings opens the parameter dialog. Show Mask runs detection (or displays a loaded mask). Add / Remove / Split Cell and Finish Mask Edit support manual QC.

This is the most powerful and configurable part of BARCC. The Mask Settings dialog provides fine-grained control over cell detection. BARCC supports two detection strategies that you can switch between at any time:

- Blob Detection (Recommended): Laplacian-of-Gaussian (blob_log) or Difference-of-Gaussian. Generally the best results on immunofluorescence images with variably bright cells. Adaptive is a checkbox on Blob/DoG, not a third radio method.
- Watershed (Legacy): The original threshold + distance transform + watershed pipeline. Retained for compatibility with older workflows. Adaptive is disabled under Watershed.

You can switch between these two methods at any time using the radio buttons. Inactive method panels are dimmed and locked. Most users should use Blob for new work, with Adaptive checked when background or density varies across the field.

NOTE

Recommended tuning order on a new image: Show Mask with defaults -> Area Tune (10 diameters) -> Smart Suggest -> optional Measure Tune precision pass (FP/TN) then recall pass (TP/FN) -> manual Add/Remove on remaining errors. For mixed packed/sparse anatomy, enable Dual Settings Mode and assign regions A/B before Smart Suggest A and B.

Detection Method

At the bottom of the Mask Settings dialog you will find a clear choice between two detection engines:

- Blob (new/recommended): Modern multi-scale blob detection using Laplacian of Gaussian. Much more robust for typical fluorescent cell images.
- Watershed (legacy): The original method based on global/local thresholding followed by watershed segmentation.

When Blob is selected, the dialog shows Blob Detection parameters (Config A; Config B as well if Dual Settings Mode is on). When Watershed is selected, the legacy Watershed parameters are shown instead. Adaptive Detection panels appear under Blob/DoG when Adaptive is checked.

Dual Settings Mode (Config A / Config B)

Check Dual Settings Mode at the top of Mask Settings to show a second Blob/Adaptive panel (Config B) to the right of Config A. The first enable copies A into B. Show Mask / Count Cells then run Config A in regions

tagged A and Config B in regions tagged B, and merge the two label maps.

- Assign a region: select it in Atlas Manager (or click it on the canvas) and press A or B. Keys are ignored while focus is in a text entry. Or right-click / ribbon: Use Config A / Use Config B.
- Unassigned regions default to Config A.
- Smart Suggest A and Smart Suggest B appear only when Dual Settings Mode and Smart Suggest's "Labeled regions only" are both on. Each recipe is fit from that group's pixels.
- Autotune shows A/B checkboxes so you can mutate one or both configs.
- Assignments travel with paint bundles. Off = one global detector.

NOTE

Use Dual Settings when one threshold cannot serve packed clusters and sparse fields on the same slice (classic example: PVN vs AHA, or left vs right with very different background). It is not a substitute for Adaptive tiles inside one config - you can use both: Adaptive on A and on B.

Adaptive Detection

Adaptive is a checkbox on Blob / LoG or DoG. It is not a third detection method. When on, BARCC estimates thresholds per tile (or per labeled region) and can run a dual-pass (sensitive + strict) then fuse detections with density-aware packing.

- `adaptive_tile_size` - tile size in pixels (typical 192-384). Smaller adapts more; larger is smoother/faster.
- `adaptive_tile_overlap` - fractional overlap 0-0.5. Higher reduces misses at tile edges.
- `adaptive_sensitivity` - global multiplier. <1 lowers tile thresholds (more cells); >1 is stricter.
- `adaptive_packing` - 0 = sparse (more free space); 1 = dense clusters; ~ 0.5 balanced.
- `adaptive_dual_pass` - 1 = sensitive + strict fused (best for mixed high/low background); 0 = single pass.
- `adaptive_region_mode` - 0 = square tiles over the whole image; 1 = one window per painted/atlas zone with its own threshold/SNR. Unlabeled tissue is not labeled. Requires zones. Then Show Mask.
- Labeled regions only (Blob Detection) - restrict Show Mask / Count Cells to painted or atlas regions even when Adaptive region mode is off. Separate from Smart Suggest's Labeled regions only checkbox.

Blob Detection Parameters (Recommended)

These parameters control the modern blob-based detector:

Blob Threshold

The most important sensitivity control. Lower values detect more (and dimmer) cells but increase false positives. Higher values are more conservative. Typical useful range: 0.05 - 0.20.

Blob Min / Max Sigma

Controls the range of cell sizes the detector will look for. Min Sigma sets the smallest detectable feature size;

Max Sigma sets the largest. For most immunofluorescence nuclei, Min Sigma around 1.5-3.0 and Max Sigma around 7-12 works well.

Blob Num Sigma

Number of scales tested between Min and Max Sigma. Higher values give finer granularity at the cost of speed. Default (12) is usually sufficient.

Blob Overlap

How much overlap is allowed between nearby blobs. Lower values reduce duplicate detections on clustered cells.

Blob Min / Max Area

Post-detection filters based on area (in pixels). Very effective at removing tiny noise blobs or huge clumps.

Blob Min Circularity

Minimum local shape circularity (0 = off, 1 = a perfect circle). Typical cells score ~0.6-0.9. Raise (0.55-0.75) to reject peanuts / merged doublets. Values above 1 (e.g. 2) are treated as 0.80. This is not the Watershed circularity box.

Peak quality filters (fewer false positives)

These gates run on candidate peaks after LoG/DoG. Hover any Mask Settings control for the same help text.

- blob_min_local_snr / blob_local_snr_outer - local signal-to-noise: (mean_core - mean_surround) / std_surround. 0 = off. Typical 1.5-3.5. Outer ring is r to r * outer (default 2.0).
- blob_bg_relative - peak minus local median on a 0-1 image. 0 = off. Try 0.08-0.18 on high background texture.
- blob_min_isotropy - radial symmetry (0 = off, 1 = perfect). Rejects edge-of-tissue and fibers bright on one side. Try 0.4-0.55.
- blob_reject_tissue_edge / blob_edge_dark_frac - reject peaks whose outer ring is partly near-black (section border). Does not treat a fully dark-field image as outside. Lower dark_frac (0.25-0.35) is stricter.
- blob_tissue_margin - integer pixels inside the OUTER slice border to reject (bright edge-line FPs). 0 = off. Try 6-12 (15-25 for thick glow). Must be a whole number.
- blob_max_elongation - max major/minor axis ratio (1 = circle). Default 3.0; 0 = off. Nuclei are typically 1-2. Rejects folds, fibers, cut-edge line.
- blob_ridge_reject / blob_ridge_thresh - Hessian ridge test (white midline / knife line). 1 = on; thresh 0.35-0.55 typical (lower drops more lines).
- blob_cavity_rim - kill zone around air-bubble bites in the section edge. Does not clear PVN/SCN next to the 3rd ventricle except peaks in the lumen. 0 = off; try 16-32.

- blob_chain_reject - 1-D line/ring suppression (ventricle wall, bubble rim, fold). 0 = off, 1 = default, 2-3 stronger. Packed 2-D clusters are kept.
- blob_cluster_recover / blob_seed_snr / blob_recover_factor - two-tier placement: bright nuclei seed a dense patch; dim neighbors within (factor x typical radius) are kept. Isolated sparse cells still pass if they look like cells. Crowded speckle without a seed is dropped. Seed SNR default ~1.15; recover factor ~3.5-4.5.
- blob_free_space - spacing between packed cells (0.05-0.95). Higher = more space (0.6-0.75 if cells look too tight). Lower = denser (0.15-0.3). Main packing knob; Adaptive packing only eases it slightly.
- blob_min_peak_intensity - normalized peak $\geq$ this (0-1). 0 = off.
- blob_exclude_border - ignore detections within N pixels of the image edge. 0 keeps border cells.
- blob_radius_scale - converts detected sigma to mask disk radius ($r \sim \sigma * \text{scale}$). Default ~1.8.

Threshold Methods

The Threshold Method controls how the image is converted to a binary (black and white) mask for cell detection. Four options are available:

- Otsu: Automatically determines the optimal global threshold using Otsu's method. Fast and effective on images with good contrast.
- Adaptive: Uses local areas to determine the threshold. Excellent for images with varying brightness across the field of view.
- Local: Similar to Adaptive but uses a different neighborhood computation. Useful when Adaptive produces too many or too few detections.
- Manual: Uses a fixed threshold value (0.0-1.0) that you specify. Provides maximum reproducibility across batches of images.

Cell Detection Parameters

These parameters control how individual cells are identified from the binary mask:

Manual Threshold

Range: 0.0 to 1.0. Only used when Threshold Method is "manual". Higher values make detection more selective (fewer cells). Lower values increase sensitivity.

Adaptive Block Size

Must be an odd integer (e.g. 51, 101, 151). Size of the local region used for adaptive thresholding. Larger values consider more surrounding area. Smaller values are more sensitive to local changes. Recommended range: 51-151 pixels.

Local Radius

Integer value (typically 5-30). Size of the neighborhood for the Local threshold method. Larger values smooth noise but may miss smaller cells.

Min Cell Size / Max Cell Size

Integer values in pixels. Define the acceptable size range for an object to be counted as a cell. Min Cell Size filters noise and small artifacts. Max Cell Size excludes clumps. Typical values depend on magnification (e.g. Min 20, Max 100-200).

Circularity Threshold

Range: 0.0 to 1.0. How circular a region must be to be accepted as a cell (1.0 = perfect circle). Higher values enforce rounder shapes. Typical value: 0.7.

Min Peak Distance

Integer value (pixels). Minimum distance required between detected cell centers. Helps prevent over-counting of touching cells. Typical values: 5-10 pixels.

Peak Min Intensity

Range: 0.0 to 1.0. Minimum brightness required for a local maximum to be considered a cell center. Higher values detect only brighter cells. Lower values detect dimmer cells. Typical starting value: 0.1.

Watershed Compactness

Range: 0.0 to 1.0. Controls how the watershed algorithm separates touching cells. Higher values favor more compact, rounder boundaries. Lower values follow intensity gradients more closely. Typical value: 0.0.

Base Multiplier & Sensitivity Range

Advanced sensitivity controls. Base Multiplier sets overall detection sensitivity (default ~1.1). Sensitivity Range controls how much the sensitivity slider can influence results (default 0.2).

Preprocessing Pipeline

A full preprocessing chain can be applied before cell detection. Each step can be enabled or disabled independently. The dialog only shows parameters relevant to the selected methods.

Background Method

Options: tophat or none. Controls removal of large-scale background variations. Tophat uses morphological operations and is generally recommended. When enabled, the Ball Radius (structural element size, default 15) controls how large-scale the background variations removed are.

Denoise Method

Options: gaussian, median, bilateral, or none. Reduces noise before detection.

- Gaussian: Applies a Gaussian blur (controlled by Gaussian Sigma, typically 0.1-5.0).
- Median: Better at preserving edges (controlled by Median Kernel size).
- Bilateral: Preserves edges while smoothing (controlled by Bilateral Sigma Color and Bilateral Sigma Space).

Contrast Method

Options: stretch, clahe, gamma, or none. Enhances contrast before detection.

- Stretch: Simple linear contrast stretching.
- CLAHE (Contrast Limited Adaptive Histogram Equalization): Local contrast enhancement. Controlled by CLAHE Kernel (typically 8-16) and CLAHE Clip Limit (typically 1.0-4.0).
- Gamma: Gamma correction. Values < 1 brighten the image; values > 1 darken it.

Enhance Method

Currently supports Unsharp Mask sharpening. Controlled by Unsharp Radius (typically 0.1-5.0) and Unsharp Amount (strength of sharpening, typically 0.1-5.0). Useful for making cell boundaries crisper.

Both manual mask editing overlays (red) and automatic cell detection masks remain visible and correctly aligned when you zoom in or out using the mouse wheel.

NOTE

Start with the default settings and adjust one parameter at a time while using "Show Mask" to preview the binary detection result. This iterative approach is the fastest way to obtain high-quality cell counts.

Autotune Panel

The Mask Settings dialog includes a convenient Autotune panel with one-click adjustments. These buttons intelligently adapt their behavior depending on whether you are using the Blob or Watershed detection method.

The available Autotune buttons are:

- More cells - Increases overall sensitivity. With Blob mode this primarily lowers the Blob Threshold and Min Sigma. With Adaptive on it also eases adaptive knobs.
- Less cells - Decreases sensitivity and raises size/shape requirements.
- Bigger cells / Smaller cells - Adjust size-related parameters (Min/Max Area or Min/Max Cell Size).
- Brighter cells / Dimmer cells - Primarily adjust intensity sensitivity (Blob Threshold or Peak Min Intensity).
- Denser packing / Sparser packing - Nudge blob_free_space (and Adaptive packing when Adaptive is

on).

- Background higher - Raise bg-relative / SNR-style gates when the field is brighter than expected.

In Dual Settings Mode, Autotune shows A/B checkboxes. Check the config(s) you want mutated. If neither is checked, Autotune asks you to pick one.

Note: The Autotune buttons are intentionally conservative. For best results on difficult images, use the new "Smart Suggest (Pre-tuning smart settings)" button instead (see below).

Export / Import Settings

You can now export your current detection and preprocessing settings as a portable .json file. This is useful for backing up configurations, sharing them with colleagues, or moving them between computers.

In the Mask Settings dialog, use the buttons at the bottom:

- "Export Settings..." - Saves your current Blob (or Watershed) configuration and preprocessing settings to a .json file of your choice.
- "Import Settings..." - Loads a previously exported .json file and applies all parameters immediately.

This is separate from the internal Presets system (which stores quick named presets locally in ~/.barc/presets.json).

Smart Suggest (Offline)

Clicking "Smart Suggest (Pre-tuning smart settings)" runs a fully local analysis on your current image and detection results. It then proposes specific parameter improvements with explanations.

- Everything runs 100% on your computer - no images or data are sent anywhere.
- Each suggestion has a checkbox. Apply All, apply checked, or Close.
- Regional diagnosis (bright vs dark tiles) and named recipes (mixed_both, high_bg_fp, low_bg_fn, recover_clusters, global over/under).
- Joint suggestions can turn Adaptive on and move dual-pass, SNR, threshold, and packing together.
- Trajectory memory after Apply (e.g. ease SNR after a strict high-BG pass).
- Prefers Area Tune / Measure Tune size bounds when those tools have been used.
- Labeled regions only - fit using only painted/atlas pixels (does not change where Show Mask draws).
- With Dual Settings Mode + Labeled regions only: Smart Suggest A and Smart Suggest B each fit their assigned region group.
- Works with Blob and Watershed; advice is context-aware from your actual data.

This tool is especially useful when Autotune is too coarse. It is the recommended way to get starting parameters for new or difficult images, after Area Tune if you have representative cells.

Area Tune

Mask Settings > Area Tune. Draw 10 independent diameter lines, one per representative cell (not across a clump). BARCC sets blob_min_area / blob_max_area to 0.7x-1.5x the mean area ($\pi * r^2$). The result is stored for the session so Smart Suggest can prefer those bounds. Measure Tune does not overwrite Area Tune size bounds when both have been used. Esc or Finish exits.

Measure Tune (TP / FP / FN / TN)

Mask Settings > Measure Tune (TP/FP/FN/TN). Label the current detection after Show Mask / Smart Suggest. Clicks snap to nearby detections. AUTO can classify a click from the image; or choose the class explicitly.

- TP (green) - real cell correctly detected.
- FP (orange) - false mark that should not be a cell.
- FN (blue) - missed cell that should have been detected.
- TN (gray) - true empty background.
- Precision pass: label FP + TN only (need ≥ 2 should-not) to tighten threshold / SNR / quality.
- Recall pass: label TP + FN only (need ≥ 2 should-detect) to recover misses.
- Full pass: both sides. Apply runs with a progress dialog (local-patch LoG, not a full-frame LoG storm).
- Detection rings stay visible while labeling. Results feed Smart Suggest.

NOTE

Expert mixed-background slices often need a precision pass, then a recall pass, then a few manual Add/Remove clicks. Do not skip Area Tune if cell size is consistent - it keeps Measure Tune from rewriting your size window.

Manual Cell Editing

No automated detector is perfect. Cell > Add Cell / Remove Cell / Split Cell / Finish Mask Edit let you correct errors before counting. Brush Settings opens automatically for add/remove. Manual edits combine with the automatic mask: (auto | add) & ~remove.

- Add Cell: click one nucleus. BARCC traces a cell-like blob at that peak; brush 1-10 tightens or expands the fill (1 = tight nucleus, 10 = more aggressive halo, 4 = traced shape). Add paint is red.
- Remove Cell: brush-erase (right-click still erases in add mode). Remove paint is yellow/gold. Detection rings stay visible under the brush, including while zooming.
- Split Cell: click a single masked blob that should be two cells. BARCC finds two intensity-weighted centers and watershed-cuts between them.
- View > Show Cell Mask toggles the red rings without re-detecting. The overlay is a persistent layer: zoom and pan no longer wipe the rings.
- Finish Mask Edit when done, then Count Cells.

Save / Load Cell Mask (cross-channel)

Cell > Save Cell Mask stores the current detection (including manual edits) as a portable .barccmask package (and a companion PNG) under output/cell_masks/. Cell > Load Cell Mask applies that mask to the current TIFF (resized if needed). While locked, Count Cells uses the loaded mask without re-running detection. Cell > Show Mask (with recalculation) unlocks and re-detects.

NOTE

Typical multi-channel flow: detect/edit cells on one channel, Save Cell Mask, Next Channel, Load Cell Mask, then Count Cells on the new channel with the same detections.

Random null cell mask

Cell > Generate Random Cell Mask creates a null distribution with the same number of cells as the ground-truth mask, at random XY locations (cell sizes shuffled). If an atlas/.catlas is loaded, counts are stratified by region: each structure keeps the same cell count as GT, placed randomly inside that structure. Display: red = ground truth, cyan = random. Count Cells does not count the random mask - only the true cell mask.

9. Axons and Nets (Intensity & PNN)

The Axons and Nets menu supports regional intensity measurement, counterstain normalization, and perineuronal (PNN) shell analysis on the loaded TIFF inside Atlas Manager regions.

Measure Region Intensities

Measures mean/median (and other stats) of image intensity inside each zone. A dialog asks whether to:

- Background subtraction: subtract the Xth-percentile intensity within each region (typical X = 5-20).
- Counterstain normalization: divide by Normalization_Factor from a file produced by Counterstain Normalization Measurement on the same atlas.

Exports under output/intensities/ as {name}_intensities.xlsx with Pre_Correction_Mean/Median and Post_Correction_Mean/Median (and detail columns). Corrected mean = $(\text{mean} - \text{BG}) / \text{factor}$ when both options are on.

Counterstain Normalization Measurement

Run this on the counterstain channel (e.g. DAPI) with the same .catlas/atlas. Saves {name}_counterstain_norm.xlsx where Normalization_Factor is the regional mean intensity (used later as a divisor for axon/PNN channels).

Perineuronal (PNN) multi-channel workflow

The recommended pipeline labels anatomy on the counterstain, detects cells on a cell channel, then measures perineuronal intensity on the quantification channel. Use the Axons and Nets and Cell menus as follows.

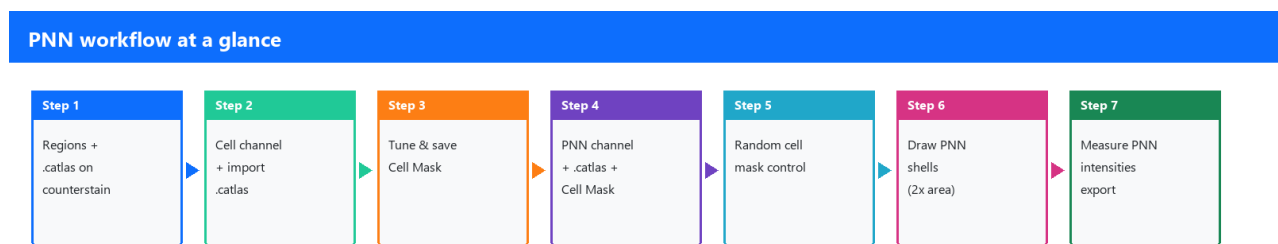


Figure. Seven-step PNN workflow at a glance (counterstain labeling through intensity export).

Step-by-step PNN procedure

Step 1 - Regions on counterstain; save .catlas

Open the counterstain image (e.g. DAPI / channel 0). Draw or import atlas regions (Allen plate, PDF atlas, or paint). Align with Fit, then Landmarks / Edge Snap / Local Refine / Crop as needed. Save Atlas Schematic as a .catlas (cropped atlas) file under output/.

Step 2 - Cell channel + import .catlas

Open the image channel that contains the cells of interest (or use Next Channel while keeping the atlas). If the atlas is not already present, Load Atlas Schematic (.catlas) onto this channel so regions match the counterstain labeling.

Step 3 - Identify cells; save Cell Mask

On the cell channel, run Cell > Show Mask / Show Mask Settings. Tune detection parameters, and use Add/Remove/Split Cell for QC. When satisfied, Cell > Save Cell Mask (.barccmask / PNG) so the same detections can be reused on other channels.

Step 4 - PNN quantification channel; import .catlas + Cell Mask

Open the channel used for perineuronal-space intensity. Import the same .catlas (if needed) and Cell > Load Cell Mask so regions and cell locations match the prior steps. The loaded cell mask is locked for Count Cells / PNN until you re-detect with Show Mask.

Step 5 - Random cell masks (control distribution)

Cell > Generate Random Cell Mask creates a matched-pair null: each random cell has the same size/shape as one true cell, at a new XY. With a .catlas loaded, placement is stratified by region. Display: red = true cells, cyan = random. (Count Cells still uses only the true mask.)

Step 6 - Draw perineuronal masks

Axons and Nets > Draw Perineuronal Masks builds shells from each cell boundary out to a disk of 2x cell area (shell = outer disk minus cell bodies). If random cells exist, random perineuronal shells are drawn too. Display: magenta = true PNN; yellow = random PNN.

Step 7 - Measure perineuronal intensities

Axons and Nets > Measure Perineuronal Intensity measures mean intensity in each shell and writes spreadsheets under output/pnn/:

- {name}_pnn_by_structure.xlsx - one row per atlas structure: True/Random Mean, SEM_Mean, Median, SEM_Median
- {name}_pnn_cells_true.xlsx - one row per true cell: Cell_Area + Perineuronal_Intensity
- {name}_pnn_cells_random.xlsx - same for random cells (when random PNN masks exist)

BARCC Perineuronal (PNN) Analysis Workflow

Multi-channel pipeline from atlas labeling through intensity export

- 1 Draw or import regions**
 Align atlas with counterstain (e.g. DAPI).
 Save a .catlas (cropped atlas) schematic.
- 2 Open channel to quantify (cells)**
 Load the image channel containing cells of interest.
 Import the .catlas onto this channel.
- 3 Identify cells**
 Run Cell Mask / tune parameters on the cell channel.
 Save the Cell Mask for reuse.
- 4 Open PNN quantification channel**
 Load the channel used for perineuronal intensity.
 Import .catlas and the saved Cell Mask.
- 5 Random control cells**
 Generate Random Cell Mask (matched-pair sizes).
 Optional stratified null by atlas region.
- 6 Draw perineuronal masks**
 Shell from cell edge to 2x cell area.
 True (magenta) and random (yellow) shells.
- 7 Measure perineuronal intensities**
 Export by-structure mean/SEM & median/SEM.
 Plus per-cell true and random spreadsheets.

BARCC v8.08+ | Axons and Nets menu | Outer shell area = 2x cell area

Figure. Detailed PNN workflow: (1) regions + .catlas on counterstain; (2) cell channel + .catlas; (3) tune and save Cell Mask; (4) PNN channel + .catlas + Cell Mask; (5) random control cells; (6) draw perineuronal shells; (7) measure and export intensities.

NOTE

Shell geometry: for cell area A , the outer disk has area $2 \times A$; the perineuronal mask is the ring between the cell body and that outer boundary. Re-draw PNN shells after regenerating random cells.

10. Counting Cells & Exporting Results

Once regions are defined and detection parameters are tuned, click "Count Cells" under the Cell menu. If Dual Settings Mode is on, counting uses Config A in A-regions and Config B in B-regions (same as Show Mask).

To review results without re-running detection, use Cell > "Show Zone Labels & Counts". This opens a table window with Zone and Cell Count columns for the current TIFF. Toggle the same menu item again (or close the window) to hide it. Yellow on-image labels also appear when counts are in memory and the option is enabled.

BARCC computes the number of detected cells within each named region. Results are saved automatically (no file dialog) under the image's output/counts/ folder (legacy flat files beside the TIFF are still detected):

- {name}.xlsx - Excel workbook with two sheets: Cell Counts (region name, cell count, area, density, ...) and Detection Parameters (every setting used, including Dual Settings / Adaptive fields when present).
- {name}_masked.tif - original image with the final cell mask (including manual Add/Remove) as a semi-transparent red overlay.
- Paint bundle / cell mask / metadata JSON as needed so Cell > Reload Last Count Session can restore the session.

Combined project spreadsheet

If you used File > Select Project Output Directory, Count Cells also appends this image to:

`{project folder}/{project name}_Counts.xlsx`

Sheet "Project Counts": first column File, then one column per unique structure name. Each image is one row. Re-running Count Cells on the same TIFF replaces that row. New structures seen later in the cohort add columns. If the workbook is locked (open in Excel), BARCC writes a CSV with the same stem.

NOTE

Per-image workbooks remain the complete record (parameters + counts). The project spreadsheet is the cohort table (one row per file). Reload Last Count Session restores the per-image mask/paint/config, not the combined workbook.

To generate the .xlsx file (instead of falling back to .csv), the following packages are required:

```
pip install openpyxl xlswriter
```

These are listed as recommended (but not strictly required) in the project's requirements.txt. Without them, results are saved as a plain CSV file.

11. Saving & Export Options

Exports are organized under the image folder's output/ directory by feature. The File Browser still finds legacy files that were saved flat in output/ or beside the TIFF.

- output/counts/ - Count Cells: {name}.xlsx (Cell Counts + Detection Parameters), _masked.tif, centroids CSV, metadata (used by Reload Last Count Session)
- Project workbook - {project folder}/{project name}_Counts.xlsx when Select Project Output Directory was used (cohort table, not under output/counts/)
- output/intensities/ - _intensities.xlsx, _counterstain_norm.xlsx
- output/pnn/ - _pnn_by_structure.xlsx, _pnn_cells_true.xlsx, _pnn_cells_random.xlsx
- output/atlas/ - .catlas schematics for multi-channel reuse
- output/cell_masks/ - .barccmask / cellmask PNG; random cell masks (_random_cellmask.png + JSON)
- output/paint/ - paint layers and .barccpaint region bundles (include Dual Settings A/B zone tags when present)
- output/flattened/ - flattened composites (TIFF + zones + paint + cell rings)

12. Keyboard Shortcuts

Shortcut	Action
Ctrl + Z	Undo last action (paint, atlas edit, etc.)
Ctrl + S	Save flattened image
Ctrl + Left	Previous image in File Browser
Ctrl + Right	Next image in File Browser
Ctrl + Shift + Right	Next uncounted image (skips excluded)
Enter	Commit crop, or painted border refit after edge drag
Esc	Cancel Landmarks / crop / Area Tune / Measure Tune
A / B	Assign selected Atlas Manager region to Dual Settings Config A or B
s (paint mode)	Toggle pen drag vs segment mode
Mouse wheel	Zoom (rings, atlas, paint stay aligned)
Alt + drag	Pan the canvas

13. Troubleshooting

Common Issues

- Cells not detected: Lower Blob Threshold, ease local SNR / bg-relative, or run Smart Suggest then a Measure Tune recall pass (TP/FN).
- Too many false positives: Raise SNR / bg-relative / isotropy; turn on ridge reject or chain reject; Measure Tune precision pass (FP/TN); Remove Cell (yellow brush).
- Packed clusters missing members: Enable blob_cluster_recover; Autotune Denser packing; Dual Settings with a dedicated Config for those zones.
- Midline / fold / bubble-rim FPs: blob_ridge_reject, blob_chain_reject, blob_cavity_rim (see Mask Settings tooltips).
- Dual Settings seems ignored: Dual Settings Mode must be on; regions must be tagged A/B (select + press A or B); then Show Mask, not only Smart Suggest.
- Atlas still far off: Fit Atlas to Image, then Landmarks with 3-6 pairs (Apply Fit), then Edge Snap in Refine mode. Use From scratch only if pose is wildly wrong. Local Refine is last.
- Edge Snap jumps the wrong way: turn on Auto tissue polarity; try Force invert for dark-field; increase tightness; Preview before Apply; Restore if needed.
- Crop box is the wrong shape: Crop checkbox on, lock aspect, pick Match TIFF or a preset before dragging. How to... in the ribbon.
- Cell rings vanish on zoom: use View > Show Cell Mask (persistent overlay in 8.10).
- Atlas drifts on zoom: Use current build (model-space img_x/img_y); re-Fit if needed.
- .catlas loads shifted: Prefer same-resolution channels; rebuild schematic after Fit; 8.08 scales layers with background size.
- Left/right structures not distinguished: Re-Reflect/stitch so names get _r/_l; too many structures may exceed uint8 bilateral split.
- Count Cells re-detects after Load Cell Mask or Reload Last Count: mask is locked until Cell > Show Mask recalculates.
- Random cells not in counts: By design - only the ground-truth cell mask is counted.
- Combined project xlsx not updating: set File > Select Project Output Directory; close the workbook in Excel if it is locked (CSV fallback).
- Excel not written: pip install openpyxl xlswriter; check output/<feature>/ is writable.

- Count Cells crashes: Use v8.06+ (TIFF deflate fix); open TIFF after the window is laid out.
- Performance is slow: Close other apps; large frames use viewer downscale - 16 GB+ RAM helps. Adaptive dual-pass on huge TIFFs is heavier than a single Blob pass.

Getting Help

For bugs or feature requests, please open an issue on the GitHub repository:

<https://github.com/LaingLab/BARCC>

Release notes for each version live in the repository root (e.g. `release-notes-v8.10.000.md`).

Thank you for using BARCC

We hope this software accelerates your research. Feedback and contributions are always welcome.